

Note the `#` prompt—you have to do that as the root user, or with `sudo`. To control services, you can use the following command:

```
#: systemctl enable unit.name # To enable a unit
#: systemctl disable unit.name # To disable a unit
#: systemctl mask unit.name # I'll explain this
#: systemctl unmask unit.name # Undo a masking
#: systemctl start unit.name # To start a unit
#: systemctl stop unit.name # To stop a unit
#: systemctl restart unit.name # To restart a unit
#: systemctl reload unit.name # To reload the config for a unit
```

The `mask` option is pretty clever. Systemd also uses symlinks, in a way. If you've disabled a unit, but another unit wants your disabled unit as a dependency, the disabled unit will still start up. What `mask` does is make the symlink for the disabled unit point to `/dev/null`. An ingenious solution, and works like a charm. And even you can't accidentally start it, unless you unmask it.

Now, there are a few commands you'll use to query and control Systemd and its units. They are:

```
#: systemctl is-enabled unit.name # Check if unit is enabled
#: systemctl status unit.name # Query the status of a unit
```

Even though the old commands are still available (they're just symlinks), you can use the following now, and they work cleanly:

```
#: systemctl reboot # Reboot the computer
#: systemctl kexec # Reboot without powering off, via kexec
#: systemctl poweroff # Halt and power down the system
#: systemctl halt # Halt the system
#: systemctl emergency # Drop to an emergency shell
#: systemctl rescue # Drop to rescue mode, equal to init 5
```

## Targets—or the replacement to runlevels

All right, this is the last section of the article before I refer you to other online resources.

Systemd uses targets to define feature points and the states of the system. A target requires a set of units, and other targets. For example, the graphical target requires the multi-user target and networking target, plus `gdm.service` and `NetworkManager.service`. Targets are also created like units, so they're simple INI files. To switch between targets, use `systemctl` (again), as shown below:

```
#: systemctl isolate target.name
```

The well-defined targets available are: `poweroff.target`, `rescue.target`, `multi-user.target`, `graphical.target`, `reboot.target`, and `emergency.target`. These are available on every Systemd machine. In addition, your distribution

may have additional targets.

It's very simple to create new targets for when your laptop is connected to a dock, and when it's not. And if you're feeling adventurous, you can hook up a few `dbus` and `udev` hooks, so that the targets are fired automatically as your laptop docks. What I've done is create a developer target that starts up PostgreSQL and Nginx, and everything that's in graphical mode.

In any case, you can set the default target by using the following command:

```
#: systemctl enable target.name
```

It's just like with enabling a unit. If you want to boot into a different target for one time, you can append a kernel parameter. For example, to boot into rescue mode, you'd append this to your kernel command line, as follows:

```
systemd.unit=rescue.target
```

## Optimisations, and more

That's it, then. And here's a note about optimisation. Systemd has a built-in read-ahead module, so to use it, you can just issue the following commands:

```
#: systemctl enable systemd-readahead-collect.service
#: systemctl enable systemd-readahead-replay.service
```

Reboot a couple of times to profile the boot process, and your boots will get even faster.

## The end

You should probably read the Arch Linux wiki article for Systemd. It has a ton of information, and it applies to all distros using Systemd, not just Arch. Among other things, it shows you how to create new units and targets. Find it at: <https://wiki.archlinux.org/index.php/Systemd>.

There's a nice GUI tool to administer Systemd—run `systemadm`.

And finally, you should read a couple of posts at Lennart Poettering's blog, for a background on the purpose of Systemd:

- <http://pointer.de/blog/projects/systemd.html>
- <http://pointer.de/blog/projects/system-update.html>
- <http://pointer.de/blog/projects/system-update-2.html>

That's it from me. 

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